In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("City_day.csv")

print("Dataset imported successfully!")
print("Shape:", df.shape)

df.head()
```

```
Dataset imported successfully!
Shape: (29531, 16)
```

Out[1]:

	City	Date	PM 2.5	PM 10	NO	NO2	
0	Ahmedabad	2015-01-01	NaN	NaN	0.92	18.22	17.
1	Ahmedabad	2015-01-02	NaN	NaN	0.97	15.69	16.
2	Ahmedabad	2015-01-03	NaN	NaN	17.40	19.30	29.
3	Ahmedabad	2015-01-04	NaN	NaN	1.70	18.48	17.
4	Ahmedabad	2015-01-05	NaN	NaN	22.10	21.42	37.

In [2]:
```
print("Dataset Information:")
df.info()

print("\nColumn Names:")
print(df.columns.tolist())

print("\nStatistical Summary:")
display(df.describe())

print("\nMissing Values:")
print(df.isnull().sum())
```

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29531 entries, 0 to 29530
Data columns (total 16 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   City        29531 non-null  object
 1   Date        29531 non-null  object
 2   PM2.5       24933 non-null  float64
 3   PM10        18391 non-null  float64
 4   NO          25949 non-null  float64
 5   NO2         25946 non-null  float64
 6   NOx         25346 non-null  float64
 7   NH3         19203 non-null  float64
 8   CO          27472 non-null  float64
 9   SO2         25677 non-null  float64
 10  O3          25509 non-null  float64
 11  Benzene     23908 non-null  float64
 12  Toluene     21490 non-null  float64
 13  Xylene      11422 non-null  float64
 14  AQI         24850 non-null  float64
 15  AQI_Bucket  24850 non-null  object
dtypes: float64(13), object(3)
memory usage: 3.6+ MB

Column Names:
['City', 'Date', 'PM2.5', 'PM10', 'NO', 'NO2', 'NOx', 'NH3', 'CO', 'SO2', 'O3
'Benzene', 'Toluene', 'Xylene', 'AQI', 'AQI_Bucket']

Statistical Summary:
```

	PM2.5	PM10	
count	24933.000000	18391.000000	25949.00000
mean	67.450578	118.127103	17.57473
std	64.661449	90.605110	22.78584
min	0.040000	0.010000	0.02000
25%	28.820000	56.255000	5.63000
50%	48.570000	95.680000	9.89000
75%	80.590000	149.745000	19.95000
max	949.990000	1000.000000	390.68000

```
Missing Values:
City              0
Date              0
PM2.5          4598
PM10          11140
NO             3582
NO2            3585
NOx            4185
NH3           10328
CO             2059
SO2            3854
O3             4022
Benzene        5623
Toluene        8041
Xylene        18109
AQI            4681
AQI_Bucket     4681
dtype: int64
```

In [3]:
```python
# Convert Date column to datetime
df["Date"] = pd.to_datetime(df["Date"], errors="coerce")

# Sort by date
df = df.sort_values("Date")

print("Relevant variables:")
print("Date, AQI, PM2.5, PM10, CO")

print(df[["Date", "AQI", "PM2.5", "PM10", "CO"]].head())
```

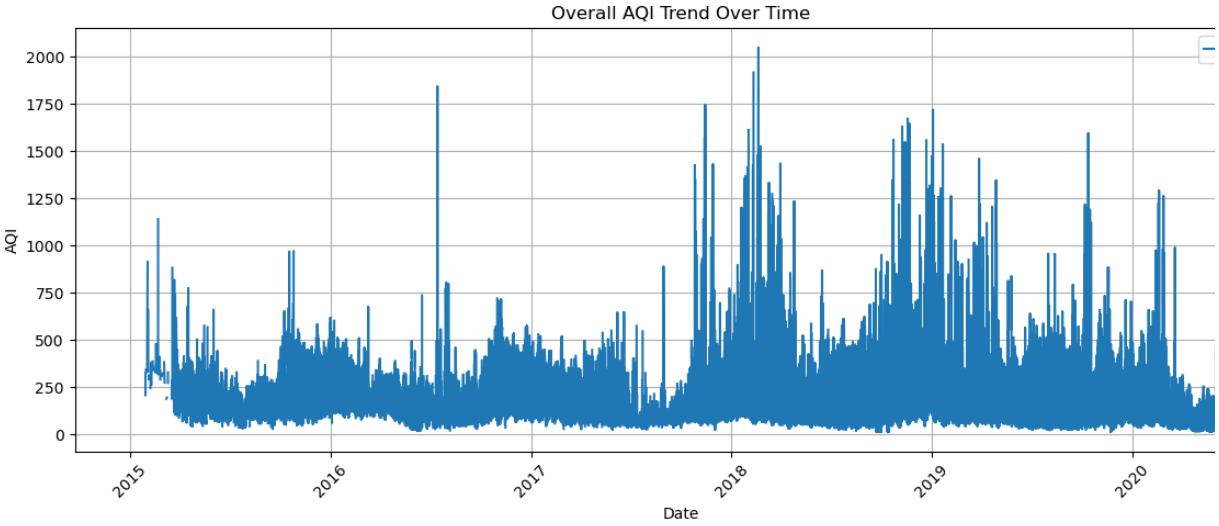
```
Relevant variables:
Date, AQI, PM2.5, PM10, CO
            Date    AQI   PM2.5    PM10     CO
0     2015-01-01    NaN     NaN     NaN   0.92
7834  2015-01-01    NaN     NaN     NaN   1.17
10229 2015-01-01  472.0  313.22  607.98  15.20
19846 2015-01-01    NaN     NaN     NaN  12.15
21855 2015-01-01    NaN     NaN     NaN   0.00
```

In [4]:
```python
plt.figure(figsize=(12, 5))

plt.plot(df["Date"], df["AQI"], label="AQI")

plt.xlabel("Date")
plt.ylabel("AQI")
plt.title("Overall AQI Trend Over Time")
plt.legend()
plt.grid(True)

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

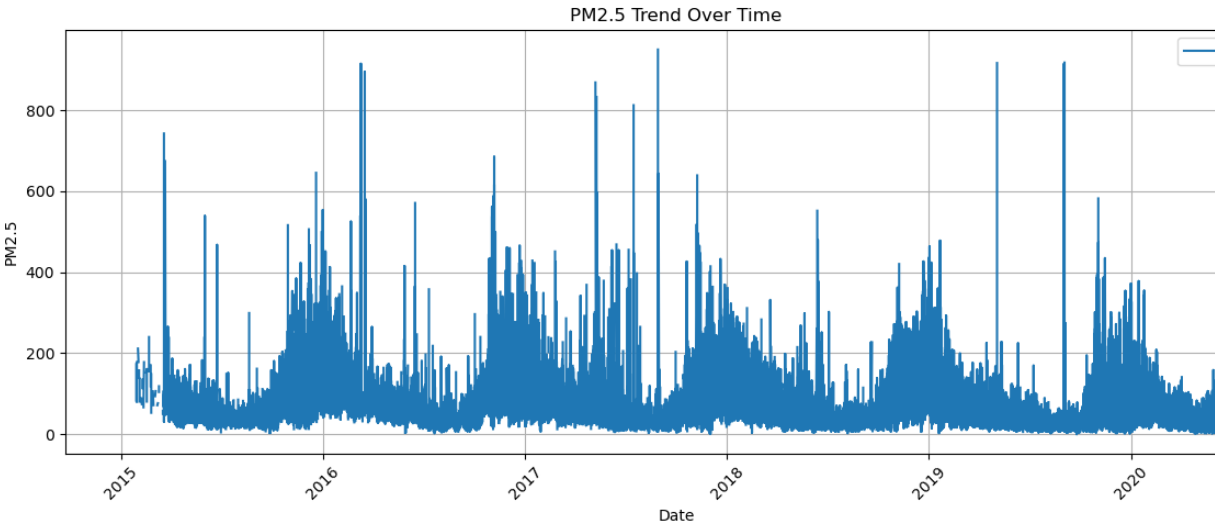


In [5]:
```
plt.figure(figsize=(12, 5))

plt.plot(df["Date"], df["PM2.5"], label="PM2.5")

plt.xlabel("Date")
plt.ylabel("PM2.5")
plt.title("PM2.5 Trend Over Time")
plt.legend()
plt.grid(True)

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



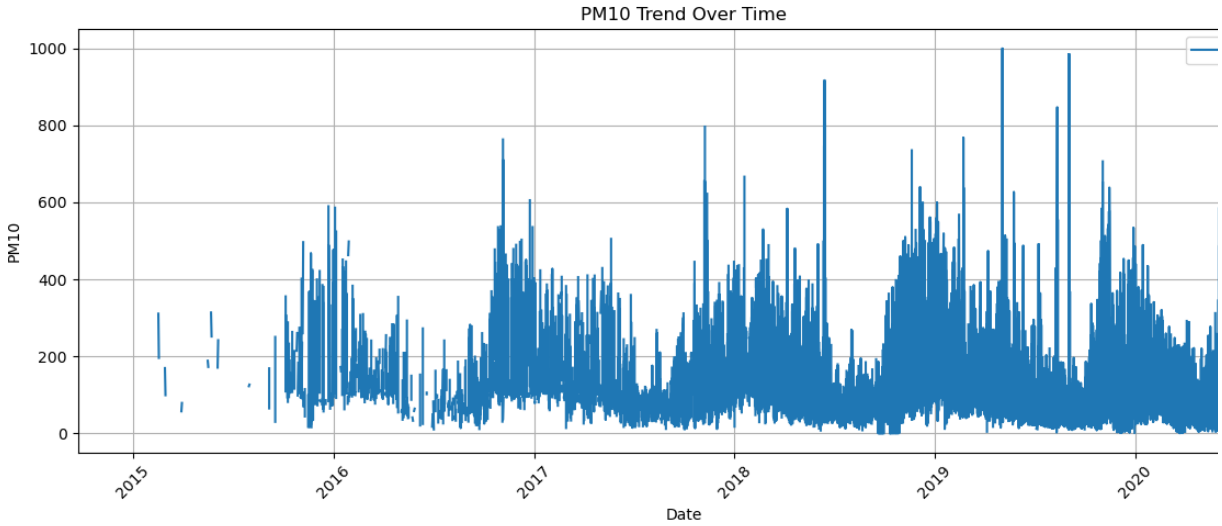
In [6]:
```
plt.figure(figsize=(12, 5))

plt.plot(df["Date"], df["PM10"], label="PM10")

plt.xlabel("Date")
plt.ylabel("PM10")
plt.title("PM10 Trend Over Time")
plt.legend()
plt.grid(True)

plt.xticks(rotation=45)
```

```
plt.tight_layout()
plt.show()
```

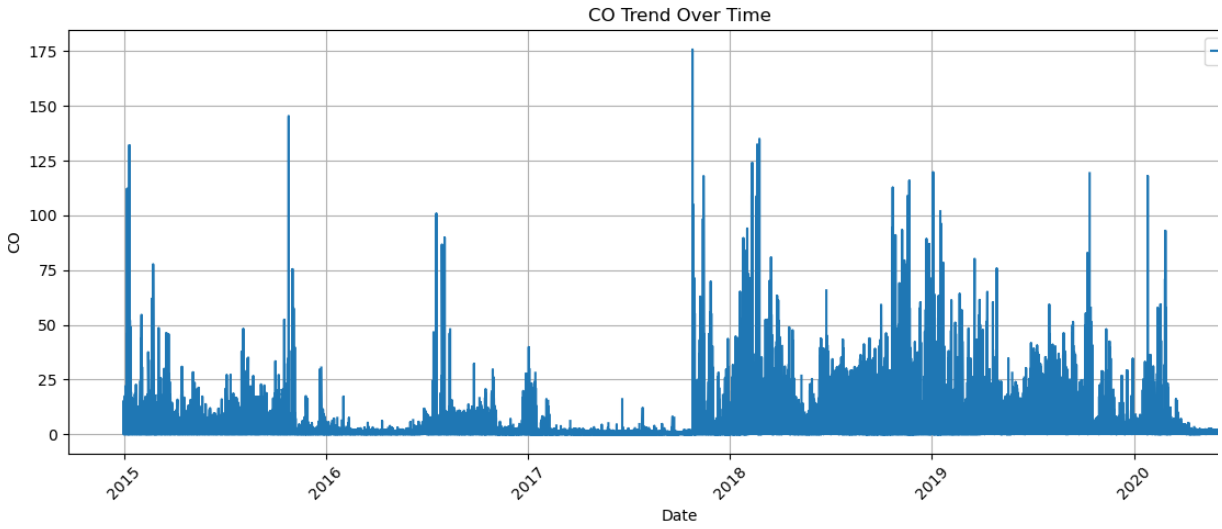


In [7]:
```
plt.figure(figsize=(12, 5))

plt.plot(df["Date"], df["CO"], label="CO")

plt.xlabel("Date")
plt.ylabel("CO")
plt.title("CO Trend Over Time")
plt.legend()
plt.grid(True)

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



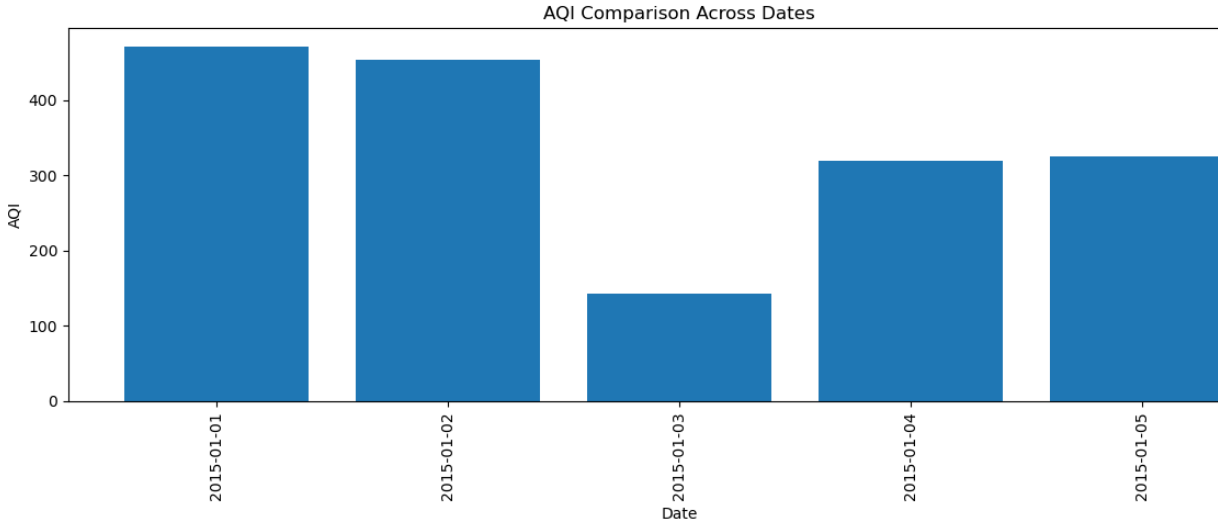
In [8]:
```
# Select a limited number of dates for better visualization
sample = df.head(30)

plt.figure(figsize=(12, 5))

plt.bar(
    sample["Date"].dt.strftime("%Y-%m-%d"),
    sample["AQI"]
)
```

```
plt.xlabel("Date")
plt.ylabel("AQI")
plt.title("AQI Comparison Across Dates")

plt.xticks(rotation=90)
plt.tight_layout()
plt.show()
```

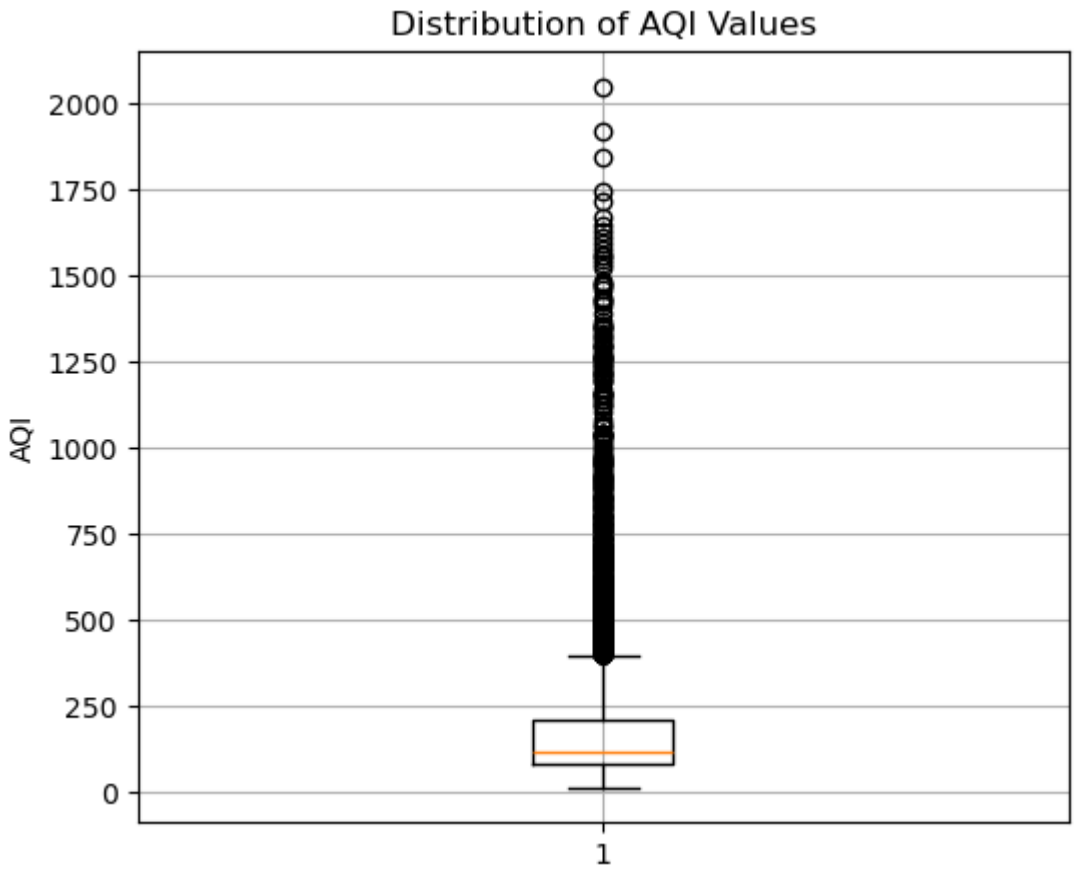


In [9]:
```
plt.figure(figsize=(6, 5))

plt.boxplot(
    df["AQI"].dropna(),
    vert=True
)

plt.ylabel("AQI")
plt.title("Distribution of AQI Values")
plt.grid(True)

plt.show()
```

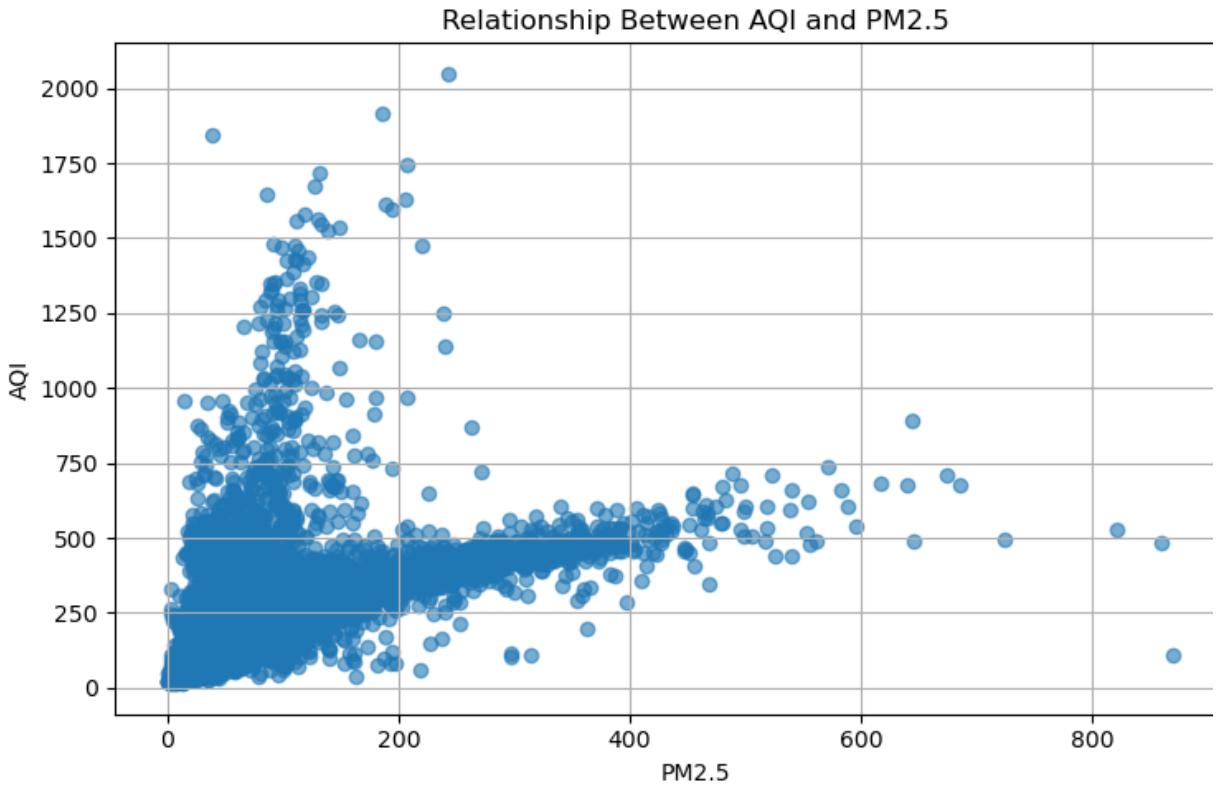


In [10]:
```
plt.figure(figsize=(8, 5))

plt.scatter(
    df["PM2.5"],
    df["AQI"],
    alpha=0.6
)

plt.xlabel("PM2.5")
plt.ylabel("AQI")
plt.title("Relationship Between AQI and PM2.5")
plt.grid(True)

plt.tight_layout()
plt.show()
```

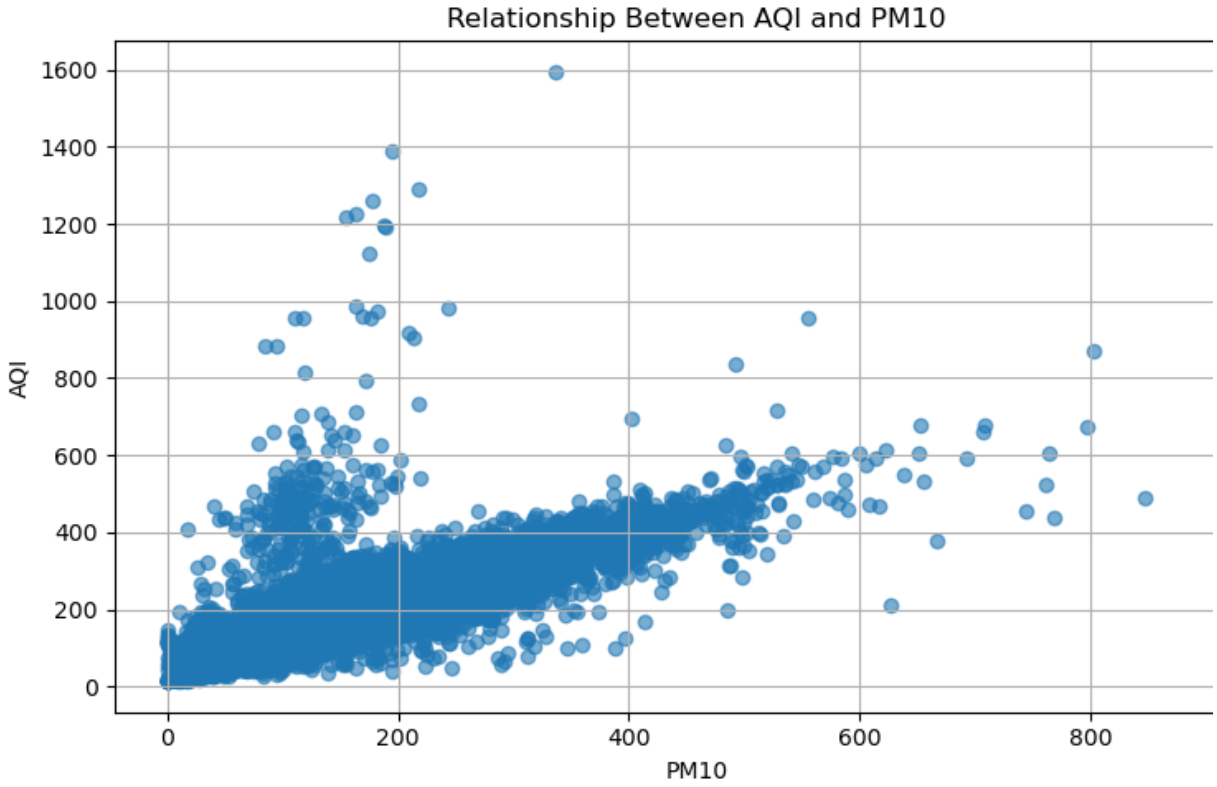


In [11]:
```
plt.figure(figsize=(8, 5))

plt.scatter(
    df["PM10"],
    df["AQI"],
    alpha=0.6
)

plt.xlabel("PM10")
plt.ylabel("AQI")
plt.title("Relationship Between AQI and PM10")
plt.grid(True)

plt.tight_layout()
plt.show()
```

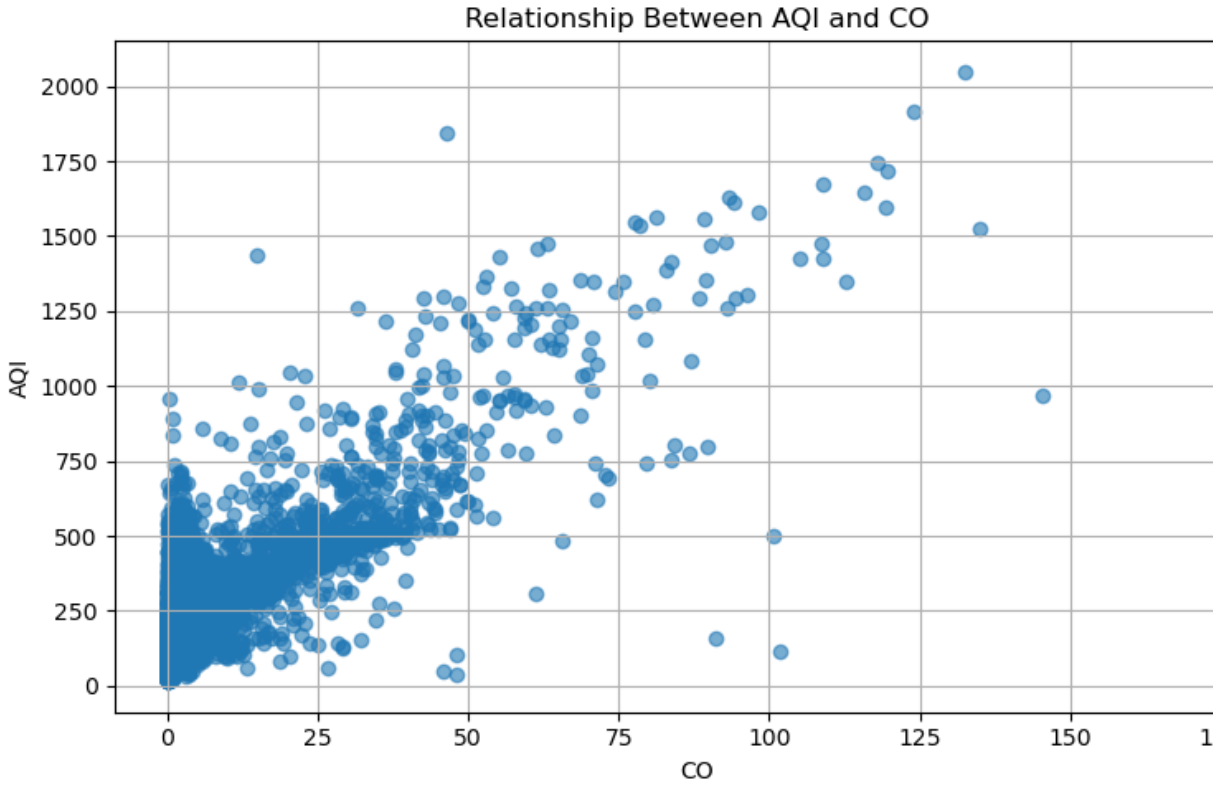


In [12]:
```
plt.figure(figsize=(8, 5))

plt.scatter(
    df["CO"],
    df["AQI"],
    alpha=0.6
)

plt.xlabel("CO")
plt.ylabel("AQI")
plt.title("Relationship Between AQI and CO")
plt.grid(True)

plt.tight_layout()
plt.show()
```



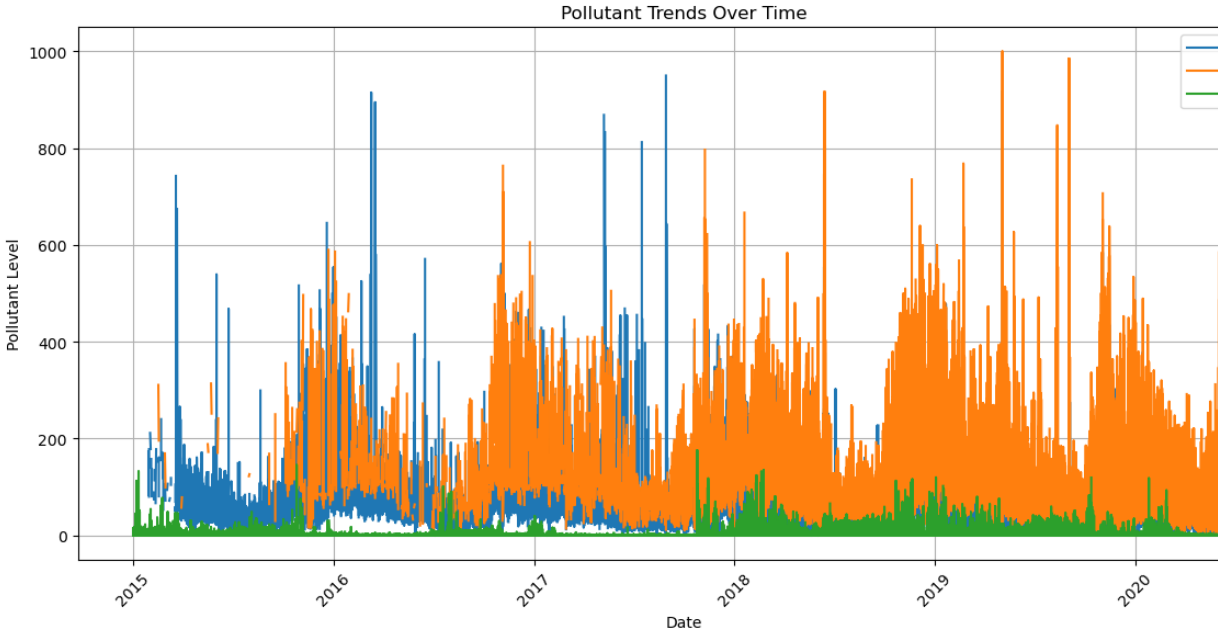
In [13]:
```
plt.figure(figsize=(12, 6))

plt.plot(df["Date"], df["PM2.5"], label="PM2.5")
plt.plot(df["Date"], df["PM10"], label="PM10")
plt.plot(df["Date"], df["CO"], label="CO")

plt.xlabel("Date")
plt.ylabel("Pollutant Level")
plt.title("Pollutant Trends Over Time")

plt.legend()
plt.grid(True)

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```


Pollutant Trends Over Time
Pollutant Level
1000
800
600
400
200
0
2015
2016
2017
2018
2019
2020
Date